YUHANG JIANG

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Homepage | Google Scholar | GitHub

Summary

- Double-degree MSc in Computer Science (University of Trento) and Data Science (SUPSI), under the EIT Manufacturing programme.
- Research intern at Huawei Pisa Research Center (multimodal agent memory) and Fondazione Bruno Kessler (embodied perception). Kaggle Competitions Expert.
- First-author research across multimodal agents, embodied AI, mechanistic interpretability of state-space models, adversarial robustness and remote sensing, with several projects run independently from problem definition to submission.
- Accepted at TMLR 2026 and a CVPR 2026 workshop; GeoSelect sets the training-free state of the art on both RRSIS-D and RISBench, reaching about 93% of a supervised specialist on RRSIS-D.

Education

EIT Manufacturing Master & Doctoral School (Double Degree)
Data Science and AI for a Competitive Manufacturing · Scholarship: full tuition waiver + 10,000 EUR/yr

- **Università di Trento (UNITN), Italy** — Sep 2023 - Mar 2026
  MSc, Computer Science · GPA 98/110 · Degree awarded Mar 2026
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI)** — Sep 2023 - Dec 2026
  Master of Engineering (MSE), Data Science · GPA 5.1/6.0 · Thesis defended Mar 2026, diploma expected Dec 2026

Anhui University (Project 211), China — Sep 2019 - Jun 2023
BEng, Intelligent Science and Technology · Average 84.34/100
Scholarships: Academic Science and Technology Scholarship, First Prize (2021); Outstanding Academic Performance, Second Prize (2022)

Research Experience

Huawei Pisa Research Center — Apr 2026 - Oct 2026
Research Intern — Pisa, Italy

- Led an independent project on memory for multimodal agents, studying how retrieved visual evidence should be selected and delivered to the model in long-horizon tasks. The resulting method reaches the best published results among memory-augmented agents on two multimodal memory benchmarks; first-author manuscript in preparation.
- LLM post-training and data generation for cockpit AI.

Fondazione Bruno Kessler (FBK) — Mar 2025 - Mar 2026
Research Intern — Trentino-Alto Adige, Italy

- Proposed Active Instance Verification, a new embodied task, and built the first offline multi-view benchmark for it in photorealistic 3D environments.
- Built an MLLM agent with attribute decomposition, confidence-weighted multi-view tracking and next-best-view planning; fine-tuned via LoRA with SFT followed by RL alignment, reaching 85.6% accuracy.

Chengdu Jiaoda Guangmang Technology Co., Ltd. — Jul 2022 - Sep 2022
Algorithm Intern — Chengdu, China

- Industrial anomaly detection: GAN-based image reconstruction and object detection for defect inspection on high-speed rail components.

Publications

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"
Transactions on Machine Learning Research (TMLR), 2026 · arXiv:2606.22516 · OpenReview
Resize-based input diversity is applied by default in transfer attacks. It reverses sign on adversarially trained surrogates, so evaluations that use it blindly overstate the robustness of the models they test.
Yuhang Jiang, Xiaojing Chen

"PInVerify: An Offline Embodied Benchmark for Active Instance Verification"
Poster, FMEA Workshop @ CVPR 2026 · Project Page · arXiv:2605.30639 · OpenReview
Proposes Active Instance Verification and builds the first benchmark for it. Active viewpoint selection gives no reliable gain over random selection; detection quality, not verification logic, binds the task.
Yuhang Jiang · Master's thesis project

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"
IEEE TGRS, under review · Project Page · arXiv:2607.03869
Reframes referring segmentation as executing an explicit spatial program, covering the positional superlative, ordinal and compositional expressions that continuous geometric fields cannot. 58.86 mIoU on RRSIS-D and 55.27 on RISBench, more than twice and about 1.7 times the best prior training-free results, with no referring supervision.
Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

Preprints and Other Publications

"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"
arXiv:2606.00930 · *Recommended for Findings by ARR meta-review*
Names the state sink, the state-space analogue of the attention sink. A near-disjoint set of heads matches the causal effect of the probe-selected specialists, and the effect dissociates by intervention surface, so a probe recovers a circuit rather than the circuit.
Yuhang Jiang, Bowen Zhang

"Task Structure Reverses Layerwise State Encoding in Sequence Models"
arXiv:2606.00926
A layerwise probe profile is evidence about the model together with what it was trained to predict, not about its architecture. Deleting the direction a probe finds does not remove what the probe read, so near-zero ablation drops do not show that a model ignores a feature.
Yuhang Jiang

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"
WRC SARA 2025 (oral) · Bowen Zhang, **Yuhang Jiang**, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"
CAIBDA 2022 (oral) · **Yuhang Jiang***, Wenping Tong

*Note: * indicates the corresponding author.*

Awards and Competitions

- **Kaggle Competitions Expert** (all solo): Silver, March Machine Learning Mania 2025 (64 / 1,727); Bronze, BirdCLEF+ 2025 (199 / 2,025); Bronze, BirdCLEF 2024 (70 / 974).
- **Mathematical Contest in Modeling (MCM), Finalist** (top 1%), 2021. Team leader; responsible for modelling and implementation. Youngest winner in the history of Anhui University.

Skills

- **Research areas:** multimodal LLMs, agent memory, LLM agents, RAG, embodied AI, mechanistic interpretability.
- **LLM / VLM methods:** SFT, DPO, GRPO / GSPO, LoRA / PEFT, multimodal inference and evaluation.
- **Frameworks and systems:** PyTorch, scikit-learn, MPI / OpenMP (HPC).
- **Programming:** Python, C/C++, Matlab, R.

Additional

- **Academic service:** Reviewer, InterpScience Workshop @ NeurIPS 2026; FMEA Workshop @ CVPR 2026.
- **Languages:** English (fluent, TOEFL 102); Chinese (native).